

SQL Code

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1
2  /* Business Analysis in SQL:
3
4  -- Business Health Overview :
5  */
6  -- Question 1: What is the overall customer churn rate?
7  -----
8  -- Total Customers
9  SELECT COUNT(*) AS total_customers
10 FROM dim_customers;
11
12 --Active Customers
13 SELECT
14     COUNT(*) AS active_customers
15 FROM fact_subscriptions
16 WHERE subscription_status = 'Active';
17
18 --Churned Customers
19 SELECT
20     COUNT(*) AS active_customers
21 FROM fact_subscriptions
22 WHERE subscription_status = 'Churned';
23
24 --Churn Rate
25 SELECT
26     COUNT(*) FILTER (WHERE subscription_status = 'Churned') AS churned_customers,
27     COUNT(*) AS total_customers,
28     ROUND(
29         COUNT(*) FILTER (WHERE subscription_status = 'Churned') * 100.0 / COUNT(*),
30         2
31     ) AS churn_rate
32 FROM fact_subscriptions;
33
34
35 --Retention Rate
36 SELECT
37     COUNT(*) FILTER (WHERE subscription_status = 'Active') AS active_customers,
38     COUNT(*) AS total_customers,
39     ROUND(
40         COUNT(*) FILTER (WHERE subscription_status = 'Active') * 100.0
41         / COUNT(*),
42         2
43     ) AS retention_rate
44 FROM fact_subscriptions;
45
46
47 -- Revenue Overview:
48 -- Question 2: How much revenue does the company generate, and how much MRR is at risk
   due to churn?
```

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49 -----
50 -----
51 /*      Total Revenue
52      Monthly Revenue
53      Revenue Lost
54      Average Revenue per Customer (ARPC)
55 */
56 --Total Revenue
57 SELECT
58     SUM(amount) AS total_revenue
59 FROM fact_payments
60 WHERE payment_status = 'paid'
61
62 --Monthly Revenue
63 SELECT
64     DATE_TRUNC('month', payment_date) AS month,
65     SUM(amount) AS monthly_revenue
66 FROM fact_payments
67 WHERE payment_status = 'paid'
68 GROUP BY DATE_TRUNC('month', payment_date)
69 ORDER BY month;
70
71 --Revenue Lost
72 SELECT
73     SUM(p.monthly_price) AS monthly_lost
74 FROM fact_subscriptions s
75 LEFT JOIN dim_plans p
76 ON s.plan_id = p.plan_id
77 WHERE s.subscription_status = 'Churned'
78
79 -----
80
81 SELECT
82     ROUND(SUM(CASE WHEN p.billing_cycle = 'Monthly' THEN monthly_price
83                  WHEN p.billing_cycle = 'Annual' THEN monthly_price /12.0
84                  END),2) AS mmr_risk
85 FROM fact_subscriptions s
86 LEFT JOIN dim_plans p
87 ON s.plan_id = p.plan_id
88 WHERE s.subscription_status = 'Churned'
89
90
91 --Average Revenue per Customer
92 SELECT
93     ROUND(SUM(p.amount)/ COUNT(s.customer_id),2) AS avg_per_cust
94 FROM fact_payments p
95 LEFT JOIN fact_subscriptions s
96 ON p.subscription_id = s.subscription_id
97 LEFT JOIN dim_customers c
98 ON s.customer_id = c.customer_id
99 WHERE p.payment_status = 'paid'
100
101 --: Customer Segmentation:
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102 --Question 3: Which customer segments have the highest churn?
103 -----
104 SELECT
105     c.segment,
106     COUNT(DISTINCT c.customer_id) AS total_customers,
107     COUNT(DISTINCT CASE
108         WHEN s.subscription_status = 'Churned'
109         THEN c.customer_id
110     END) AS churned_customers,
111     ROUND(
112         COUNT(DISTINCT CASE
113             WHEN s.subscription_status = 'Churned'
114             THEN c.customer_id
115         END) * 100.0 /
116         COUNT(DISTINCT c.customer_id),
117         2
118     ) AS churn_rate
119 FROM dim_customers c
120 JOIN fact_subscriptions s
121 ON c.customer_id = s.customer_id
122 GROUP BY c.segment
123 ORDER BY churn_rate DESC;
124
125 --Subscription Analysis:
126 --Question 4: Which subscription plans have the highest churn and which generate the
127 most revenue?
128 -----
129 WITH CTE AS(
130     SELECT
131         p.plan_name,
132         COUNT(CASE WHEN subscription_status = 'Churned' THEN s.subscription_id END) AS
highest_churn ,
133         SUM(CASE
134             WHEN fp.payment_status = 'paid' THEN fp.amount ELSE 0 END) AS
most_revenue,
135         COUNT(DISTINCT s.subscription_id) AS total_subscription
136 FROM fact_subscriptions s
137 LEFT JOIN dim_plans p
138 ON s.plan_id = p.plan_id
139 LEFT JOIN fact_payments fp
140 ON s.subscription_id = fp.subscription_id
141 GROUP BY p.plan_name
142 HAVING p.plan_name IS NOT NULL
143 ORDER BY highest_churn DESC
144 )
145 SELECT
146     plan_name,
147     highest_churn,
148     ROUND((highest_churn * 100.0 / total_subscription),2) AS churn_rate,
149     most_revenue
150 FROM CTE
151

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152 --Customer Engagement
153 --Question 5: Does product usage affect churn?
154 -----
155 WITH customer_usage AS(
156 SELECT
157     customer_id,
158     SUM(login_count) AS total_logins
159 FROM fact_usage
160 GROUP BY customer_id
161 )
162 ,login_bucket AS(
163 SELECT
164     customer_id,
165     total_logins,
166     CASE
167         WHEN total_logins <= 50 THEN 'Low (0-50)'
168         WHEN total_logins <= 100 THEN 'Medium (51-100)'
169         WHEN total_logins <= 150 THEN 'High (101-150)'
170         ELSE 'Power User (150+)'
171     END AS login_bucket
172 FROM customer_usage
173 ),
174 churn_rates AS(
175 SELECT
176     lb.login_bucket,
177     COUNT(DISTINCT s.subscription_id) AS total_subscription,
178     COUNT(CASE WHEN s.subscription_status = 'Churned' THEN s.subscription_id
179     END) AS churned_subscription
180
181 FROM login_bucket lb
182 LEFT JOIN fact_subscriptions s
183 ON lb.customer_id = s.customer_id
184 GROUP BY lb.login_bucket
185 )
186 SELECT
187     login_bucket,
188     total_subscription,
189     churned_subscription,
190     ROUND((churned_subscription * 100.0 / total_subscription ),2) AS churn_rate
191 FROM churn_rates
192
193
194 -- Support Analysis
195 --Question 6: Does customer support impact churn?
196 -----
197 WITH support_summary AS (
198     SELECT
199         s.customer_id,
200         s.subscription_id,
201         s.subscription_status,
202         COUNT(fs.ticket_id) AS total_tickets
203 FROM fact_subscriptions s
204 LEFT JOIN fact_supports fs
205     ON s.customer_id = fs.customer_id
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206         GROUP BY
207             s.customer_id,
208             s.subscription_id,
209             s.subscription_status
210     ),
211     ticket_bucket AS (
212         SELECT
213             customer_id,
214             subscription_id,
215             subscription_status,
216             total_tickets,
217             CASE
218                 WHEN total_tickets = 0 THEN '0 Tickets'
219                 WHEN total_tickets = 1 THEN '1 Ticket'
220                 WHEN total_tickets BETWEEN 2 AND 3 THEN '2-3 Tickets'
221                 WHEN total_tickets BETWEEN 4 AND 5 THEN '4-5 Tickets'
222                 ELSE '6+ Tickets'
223             END AS ticket_bucket
224         FROM support_summary
225     )
226     SELECT
227         ticket_bucket,
228         COUNT(DISTINCT subscription_id) AS total_subscriptions,
229         COUNT(
230             DISTINCT CASE
231                 WHEN subscription_status = 'Churned'
232                 THEN subscription_id
233             END
234         ) AS churned_subscriptions,
235         ROUND(
236             COUNT(
237                 DISTINCT CASE
238                     WHEN subscription_status = 'Churned'
239                     THEN subscription_id
240                 END
241             ) * 100.0
242             / COUNT(DISTINCT subscription_id),
243             2
244         ) AS churn_rate
245
246     FROM ticket_bucket
247     GROUP BY ticket_bucket
248     ORDER BY
249         CASE ticket_bucket
250             WHEN '0 Tickets' THEN 1
251             WHEN '1 Ticket' THEN 2
252             WHEN '2-3 Tickets' THEN 3
253             WHEN '4-5 Tickets' THEN 4
254             WHEN '6+ Tickets' THEN 5
255         END;
256
257
258
259     -- Payment Analysis:
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260 -- Question 7: Are payment issues associated with churn?
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263
264 WITH payment_summary AS (
265     SELECT
266         subscription_id,
267         COUNT(payment_id) AS total_payments,
268         COUNT(
269             CASE
270                 WHEN payment_status = 'failed'
271                 THEN payment_id
272             END
273         ) AS failed_payments
274     FROM fact_payments
275     GROUP BY subscription_id
276 ),
277
278 payment_bucket AS (
279     SELECT
280         subscription_id,
281         total_payments,
282         failed_payments,
283         CASE
284             WHEN failed_payments = 0 THEN 'No Failed Payments'
285             WHEN failed_payments = 1 THEN '1 Failed Payment'
286             ELSE '2+ Failed Payments'
287         END AS payment_bucket
288     FROM payment_summary
289 )
290
291 SELECT
292     pb.payment_bucket,
293
294     COUNT(DISTINCT s.subscription_id) AS total_subscriptions,
295
296     COUNT(
297         DISTINCT CASE
298             WHEN s.subscription_status = 'Churned'
299             THEN s.subscription_id
300         END
301     ) AS churned_subscriptions,
302
303     ROUND(
304         COUNT(
305             DISTINCT CASE
306                 WHEN s.subscription_status = 'Churned'
307                 THEN s.subscription_id
308             END
309         ) * 100.0 /
310         COUNT(DISTINCT s.subscription_id),
311         2
312     ) AS churn_rate
313
```

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314 FROM payment_bucket pb
315 JOIN fact_subscriptions s
316     ON pb.subscription_id = s.subscription_id
317
318 GROUP BY pb.payment_bucket
319
320 ORDER BY
321     CASE pb.payment_bucket
322         WHEN 'No Failed Payments' THEN 1
323         WHEN '1 Failed Payment' THEN 2
324         WHEN '2+ Failed Payments' THEN 3
325     END;
326
327 -----
328 SELECT
329     p.payment_method,
330     COUNT(DISTINCT s.subscription_id) AS total_subscriptions,
331     COUNT(
332         DISTINCT CASE
333             WHEN s.subscription_status = 'Churned'
334             THEN s.subscription_id
335         END
336     ) AS churned_subscriptions,
337     ROUND(
338         COUNT(
339             DISTINCT CASE
340                 WHEN s.subscription_status = 'Churned'
341                 THEN s.subscription_id
342             END
343         ) * 100.0 /
344         COUNT(DISTINCT s.subscription_id),
345         2
346     ) AS churn_rate
347 FROM fact_payments p
348 JOIN fact_subscriptions s
349     ON p.subscription_id = s.subscription_id
350 GROUP BY p.payment_method
351 ORDER BY churn_rate DESC;
352 -----
353
354 SELECT
355     payment_method,
356     SUM(amount) AS total_revenue
357 FROM fact_payments
358 WHERE payment_status = 'paid'
359 GROUP BY payment_method
360 ORDER BY total_revenue DESC;
361
362
363 -- Time-Based Analysis:
364 -- Question 8: How have subscriptions, churn, and revenue changed over time?
365 -----
366 -- New customer:
367 SELECT

```

```
368         DATE_TRUNC('month',signup_date)::DATE AS month_date,
369         COUNT(customer_id) AS new_customer
370     FROM dim_customers
371     GROUP BY 1
372     ORDER BY month_date
373
374     -- Monthly Churn customer:
375     SELECT
376         DATE_TRUNC('month',end_date)::DATE AS months,
377         COUNT(subscription_id) AS churn_customer
378     FROM fact_subscriptions
379     WHERE subscription_status = 'Churned'
380     GROUP BY 1
381     ORDER BY churn_customer DESC
382
383     -- Monthly Revenue:
384     SELECT
385         DATE_TRUNC('month',payment_date)::DATE AS months,
386         SUM(amount) AS monthly_revenue
387     FROM fact_payments
388     WHERE payment_status = 'paid'
389     GROUP BY 1
390     ORDER BY monthly_revenue DESC
391
392     -- Monthly Active Subscriptions:
393     SELECT
394         DATE_TRUNC('month', start_date)::DATE AS month,
395         COUNT(subscription_id) AS new_subscriptions
396     FROM fact_subscriptions
397     WHERE subscription_status = 'Active'
398     GROUP BY 1
399     ORDER BY 1;
400
401
402     -- High-Value Customer Analysis
403     -- Question 9: What are the characteristics of churned customers across revenue,
404     -- product usage, and support history?
405     -----
406     -----
407     WITH revenue_cte AS (
408         SELECT
409             s.customer_id,
410             SUM(
411                 CASE
412                     WHEN p.payment_status = 'paid' THEN p.amount
413                     ELSE 0
414                 END
415             ) AS lifetime_revenue
416         FROM fact_payments p
417         JOIN fact_subscriptions s
418             ON p.subscription_id = s.subscription_id
419         GROUP BY s.customer_id
420     ),
```



```
420 usage_cte AS (  
421     SELECT  
422         customer_id,  
423         SUM(login_count) AS total_logins  
424     FROM fact_usage  
425     GROUP BY customer_id  
426 ),  
427  
428 support_cte AS (  
429     SELECT  
430         customer_id,  
431         COUNT(ticket_id) AS total_tickets  
432     FROM fact_supports  
433     GROUP BY customer_id  
434 ),  
435  
436 subscription_cte AS (  
437     SELECT  
438         s.customer_id,  
439         dp.plan_name,  
440         s.subscription_status,  
441         ROW_NUMBER() OVER (  
442             PARTITION BY s.customer_id  
443             ORDER BY s.start_date DESC  
444         ) AS rn  
445     FROM fact_subscriptions s  
446     JOIN dim_plans dp  
447         ON s.plan_id = dp.plan_id  
448 )  
449  
450 SELECT  
451     sc.customer_id,  
452     sc.plan_name AS current_plan,  
453     sc.subscription_status,  
454     COALESCE(r.lifetime_revenue, 0) AS lifetime_revenue,  
455     COALESCE(u.total_logins, 0) AS total_logins,  
456     COALESCE(sp.total_tickets, 0) AS total_tickets  
457 FROM subscription_cte sc  
458 LEFT JOIN revenue_cte r  
459     ON sc.customer_id = r.customer_id  
460 LEFT JOIN usage_cte u  
461     ON sc.customer_id = u.customer_id  
462 LEFT JOIN support_cte sp  
463     ON sc.customer_id = sp.customer_id  
464 WHERE sc.rn = 1  
465     AND sc.subscription_status = 'Churned'  
466 ORDER BY lifetime_revenue DESC,  
467         total_logins DESC;  
468  
469  
470  
471  
472
```

473	
-----	--